

Lab 11.5 - Test Suite Summary Document

Module 11: Testing & Debugging in Flutter | Taskly App

1. Test Suite Summary

The Taskly test suite contains a total of **11 tests** across four distinct testing categories, achieving a **100% pass rate**.

Metric	Value
Total Tests Executed	11
Tests Passed	11
Tests Failed	0
Pass Rate	100%
Average Execution Time	~3.5 seconds

2. Test Classifications & Behaviors Validated

2.1 Unit Tests (5 Tests)

- **Task Model (2 Tests):** Validates that the default task status is incomplete (false) and that calling toggle() correctly switches it true ↔ false.
- **TaskRepository (3 Tests):** Validates that adding, deleting, and updating tasks modify the internal collections correctly and notify UI listeners.

2.2 Widget Tests (3 Tests)

- **Empty State:** Verifies the prompt 'No tasks yet. Add one!' is rendered when no tasks are loaded.
- **Task Addition:** Verifies that typing text into the TextField and clicking the 'Add' button adds the item to the screen and removes the empty state.
- **Multiple Tasks:** Verifies that multiple added tasks render simultaneously in the list view.

2.3 Navigation Tests (1 Test)

- Verifies that tapping on a task tile triggers navigation to the task details page, opens with the correct header 'Task Detail', and exposes the detailTextField with Key 'detailTitleField'.

2.4 Integration Tests (2 Tests)

- Simulates the end-to-end user lifecycle: adding a task ('Original title'), tapping it to open the details view, typing a new name ('Updated title'), saving it, and verifying that the updated title replaces the original title in the main list view.

3. Known Limitations & Recommendations

- **Ephemeral State:** The task list is volatile and reset upon restarts. Persistence via Hive or SQLite is recommended.
- **Mocked Timings:** Widget and integration tests run on standard headless flutter mock timings, which may not capture frame rendering lags.
- **No Server Integration:** Syncing processes are not verified as the app operates purely locally.